Musiiwa Thandabantu Data Analytics Google BigQuery Project

Question 1

```
---Question1: WHERE Clause
---Filter all transactions that occurred in the year 2023.
---Expected Output: All columns
SELECT *
FROM `bright-tv-493010.Retail.retail_sales`
WHERE Date BETWEEN '2023-01-01' AND '2023-12-31';
```

Syntax error: Expected end of input but got keyword SELECT at [7:1]

Using on-demand processing quota

Query results

+ Create conversation Save results Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	191	2023-10-18	CUST191	Male	64	Beauty	1	25	
2	204	2023-09-28	CUST204	Male	39	Beauty	1	25	
3	230	2023-04-23	CUST230	Male	54	Beauty	1	25	
4	232	2023-02-06	CUST232	Female	43	Beauty	1	25	
5	309	2023-12-23	CUST309	Female	26	Beauty	1	25	
6	310	2023-10-12	CUST310	Female	28	Beauty	1	25	
7	363	2023-06-03	CUST363	Male	64	Beauty	1	25	
8	371	2023-02-21	CUST371	Female	20	Beauty	1	25	
9	397	2023-03-10	CUST397	Female	30	Beauty	1	25	

Question 2

```
---Question 2: Filtering + Conditions
---Display all transactions where the Total Amount is more than the average Total Amount of the entire dataset.
---Expected Output: All columns

SELECT *
FROM `bright-tv-493010.Retail.retail_sales`
WHERE `Total Amount` > (SELECT AVG (`Total Amount`)
FROM `bright-tv-493010.Retail.retail_sales`);
```

Syntax error: Expected end of input but got keyword SELECT at [7:1]

Query results

+ Create conversation Save results Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	21	2023-01-14	CUST021	Female	50	Beauty	1	500	
2	28	2023-04-23	CUST028	Female	43	Beauty	1	500	
3	128	2023-07-05	CUST128	Male	25	Beauty	1	500	
4	220	2023-03-03	CUST220	Male	64	Beauty	1	500	
5	238	2023-01-17	CUST238	Female	39	Beauty	1	500	
6	364	2023-08-23	CUST364	Female	19	Beauty	1	500	
7	408	2023-04-15	CUST408	Female	64	Beauty	1	500	
8	537	2023-06-03	CUST537	Female	21	Beauty	1	500	
9	539	2023-06-08	CUST539	Male	25	Beauty	1	500	

Question 3

```
---Question 3: Aggregate Functions
--Calculate the total revenue (sum of Total Amount).
---Expected Output: Total_Revenue

SELECT SUM(`Total Amount`) AS Total_Revenue
FROM `bright-tv-493010.Retail.retail_sales`;
```

Query completed

Query results

+ Create conversation

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Total_Revenue
1	456000

Question 4

```
---Question 4: DISTINCT
---Display all distinct Product Categories in the dataset.
---Expected Output: Product_Category

SELECT DISTINCT(`Product Category`)
FROM `bright-tv-493010.Retail.retail_sales`;
```

Syntax error: Expected end of input but got keyword SELECT at [7:1]

Query results

+ Create conversation | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product Category
1	Beauty
2	Clothing
3	Electronics

Question 5

```
---Question 5 GROUP BY
---For each Product Category, calculate the total quantity sold.
---Expected Output: Product_Category, Total_Quantity

SELEC  Open resource side-by-side. (cmd + click)  Total_Quantity
FROM `bright-tv-493010.Retail.retail_sales`
GROUP BY `Product Category`;
```

Syntax error: Expected end of input but got keyword SELECT at [7:1]

Query results

+ Create conversation Save

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product Category	Total_Quantity
1	Beauty	771
2	Clothing	894
3	Electronics	849

Question 6

```
---Question 6: CASE Statement
---Create a column called Age_Group that classifies customers as:
- 'Youth' (<30)
- 'Adult' (30-59)
- 'Senior' (60+)
---Expected Output: Customer_ID, Age, Age_Group

SELECT 'Customer ID', Age,
CASE
  WHEN Age <30 THEN 'Youth'
  WHEN Age BETWEEN 30 AND 59 THEN 'Adult'
  ELSE 'Senior'
  END AS Age_Group
FROM `bright-tv-493010.Retail.retail_sales`;
```

Syntax error: Expected end of input but got keyword SELECT at [7:1]

Query results

+ Create conversation Save results Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	f0_	Age	Age_Group
1	Customer ID	64	Senior
2	Customer ID	39	Adult
3	Customer ID	54	Adult
4	Customer ID	43	Adult
5	Customer ID	26	Youth
6	Customer ID	28	Youth
7	Customer ID	64	Senior
8	Customer ID	20	Youth

Question 7

```sql
---Question 7: Conditional Aggregation
---For each Gender, count how many high-value transactions occurred (where Total Amount > 500).
---Expected Output: Gender, High_Value_Transactions

SELECT Gender,COUNT('Transaction ID') AS High_Value_Transactions
FROM `bright-tv-493010.Retail.retail_sales`
WHERE `Total Amount` >500
GROUP BY Gender;
```

Syntax error: Expected end of input but got keyword SELECT at [7:1]

Query results

+ Create conversation

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Gender	High_Value_Tran...
1	Female	155
2	Male	144

Question 8

```sql
---Question 8 HAVING Clause
---For each Product Category, show only those categories where the total revenue exceeds 5,000.
---Expected Output: Product_Category, Total_Revenue
SELECT `Product Category`, SUM(`Total Amount`) AS Total_Revenue
FROM `bright-tv-493010.Retail.retail_sales`
GROUP BY `Product Category`
HAVING SUM(`Total Amount`) >5000;
```

Syntax error: Expected end of input but got keyword SELECT at [7:1]

Query results

+ Create conversation

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

Question 9

```sql
---Question 9: Calculated Fields
---Display a new column called Unit_Cost_Category that labels a transaction as:
--- 'Cheap' if Price per Unit < 50
---'Moderate' if Price per Unit between 50 and 200
--'Expensive' if Price per Unit > 200
---Expected Output: Transaction_ID, Price_per_Unit, Unit_Cost_Category

SELECT `Transaction ID`, `Price per unit`,
CASE
  WHEN `Price per Unit`< 50 THEN 'Cheap'
  WHEN `Price per Unit`BETWEEN 50 AND 200 THEN 'Moderate'
  WHEN `Price per Unit`> 200 THEN 'Expensive'
  END AS `Unit_Cost Category`
FROM `bright-tv-493010.Retail.retail_sales`;
```

Syntax error: Expected end of input but got keyword SELECT at [7:1]

Query results

+ Create conversa

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Price per unit	Unit_Cost Category
1	191	25	Cheap
2	204	25	Cheap
3	230	25	Cheap
4	232	25	Cheap
5	309	25	Cheap
6	310	25	Cheap
7	363	25	Cheap

Results per page

Question 10

```sql
---Question 10: Combining WHERE + CASE
---Display all transactions from customers aged 40 or older and add a column Spending_Level
---showing:
 'High' if Total Amount > 1000
 'Low' otherwise
---Expected Output: Customer_ID, Age, Total_Amount, Spending_Level

SELECT `Customer ID`, Age, `Total Amount`,
CASE
  WHEN `Total Amount` >1000 THEN 'High'
  ELSE 'Low'
  END AS Spending_Level
  FROM `bright-tv-493010.Retail.retail_sales`
  WHERE Age >=40;
```

Syntax error: Expected end of input but got keyword SELECT at [7:1]

Query results

+ Create conversa

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Customer ID	Age	Total Amount	Spending_Level
1	CUST191	64	25	Low
2	CUST230	54	25	Low
3	CUST232	43	25	Low
4	CUST363	64	25	Low
5	CUST454	46	25	Low
6	CUST512	57	25	Low
7	CUST791	51	25	Low